

Parameter	N-cycle	S-cycle	I-cycle	C-cycle
Full name	Non-sequential cycle	Sequential cycle	Internal cycle	Coprocessor register-transfer cycle
External memory transfer	Yes	Yes	No	No ordinary memory transfer
Purpose	First/new memory access	Following sequential memory access	Internal CPU processing	ARM–coprocessor register transfer
Address relationship	Not sequential with previous transfer	Sequential with previous transfer	No valid external memory transfer required	Not an ordinary memory address transfer
nMREQ	LOW	LOW	HIGH	HIGH
SEQ	LOW	HIGH	LOW	HIGH
Memory controller	Must perform a new access	Can exploit sequential access	Memory not requested	Memory not requested
Typical example	First fetch after branch	Consecutive instruction fetch	Internal cycle of multi-cycle instruction	Coprocessor register transfer